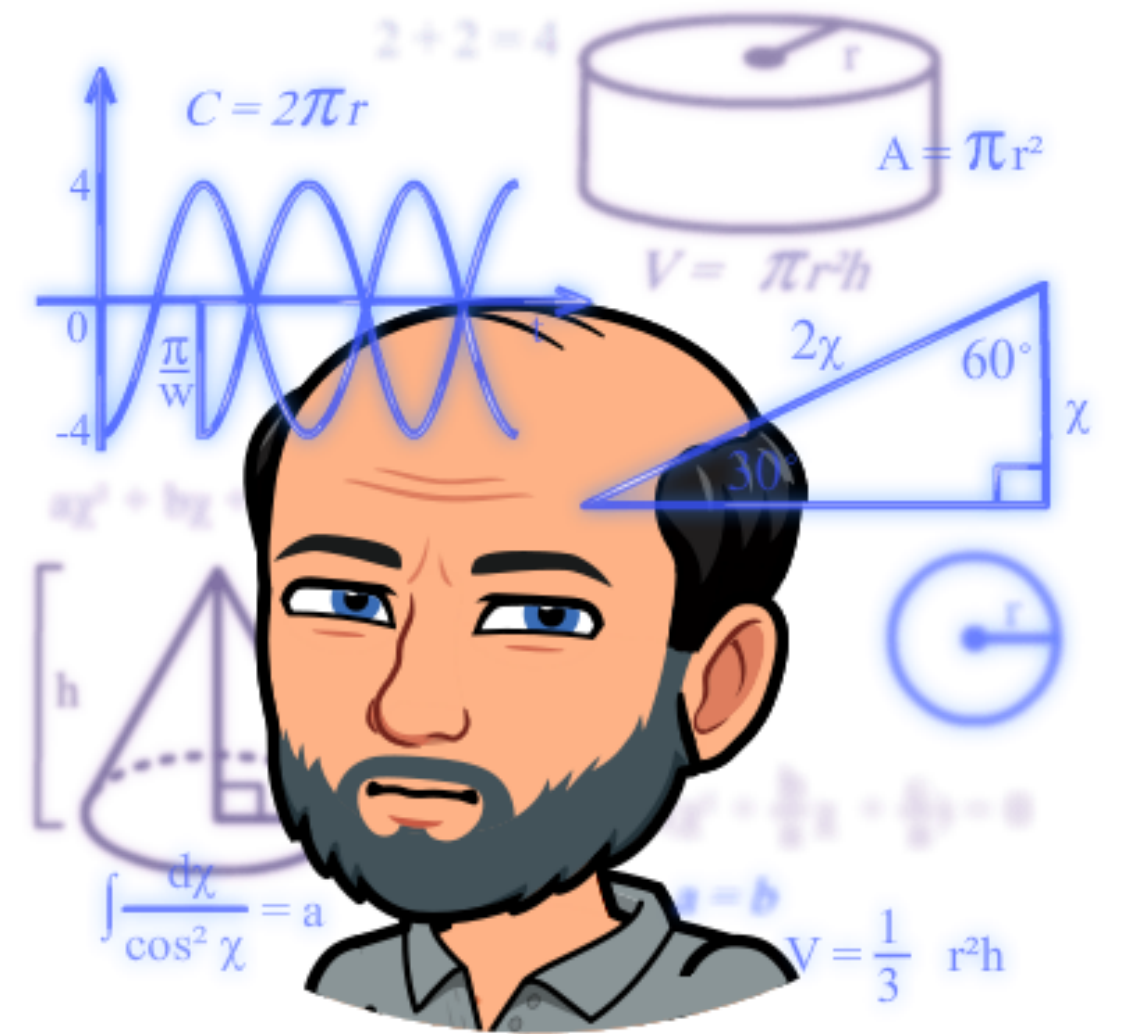


Window functions in SQL Server

- brush up your skills



Thomas Hütter

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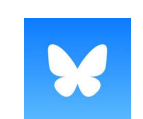
Window functions in SQL Server

Thomas Hütter, Diplom-Betriebswirt

- Developer for ERP apps, SQL scripts and BI stuff
- Worked at consultancies, ISVs, end user companies
- 1995: SQL Server, 2014: R, 2020: Power*, 2024: Arduino
- Speaker at SQL / data / dev events around Europe



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Agenda

- Introduction: window functions
- Aggregate functions
- Window frames
- Ranking functions
- Statistical / analytical functions
- Offset functions
- Specials
- Round-up, resources, learning

Introduction: window functions

- „In SQL, a window function or analytic function is a function which uses values from one or multiple rows to return a value for each row.
(This contrasts with an aggregate function, which returns a single value for multiple rows.)
Window functions have an OVER clause; any function without an OVER clause is not a window function, but rather an aggregate or single-row (scalar) function.“ *1)
- „A window function in SQL is a type of function that allows us to perform calculations across a specific set of rows related to the current row. These calculations happen within a defined window of data, and they are particularly useful for aggregates, rankings, and cumulative totals without altering the dataset.“ *2)
- A window function applies additional filtering to a query, which can be defined by partitioning, ordering, and framing clauses.
Contrary to a WHERE clause, this works on each row individually.

*1: [https://en.wikipedia.org/wiki/Window_function_\(SQL\)](https://en.wikipedia.org/wiki/Window_function_(SQL)), *2: <https://www.geeksforgeeks.org/window-functions-in-sql>

Introduction: window functions

- **General form:**

```
SELECT <column_1>, <column_2>,  
<window_function> OVER ( PARTITION BY <...> | ORDER BY <...> | <window_frame>) <column_alias>  
FROM <table_name> ...
```

- **<window_frame>:**

Rows | Range [| Groups] BETWEEN <lower_bound> AND <upper_bound> [<window frame exclusion>]
bounds: Unbounded preceding | N preceding | Current row | N following | Unbounded following

- **Named window definition (SQL Server 2022):**

```
SELECT <column_1>, <column_2>,  
<window_function_1>() OVER <window_name> <column_alias_1>  
<window_function_2>() OVER <window_name> <column_alias_2>  
FROM <table_name> ...  
WINDOW <window_name> AS (PARTITION BY <...> | ORDER BY <...> | <window_frame>)
```


Aggregate functions

The usual suspects:

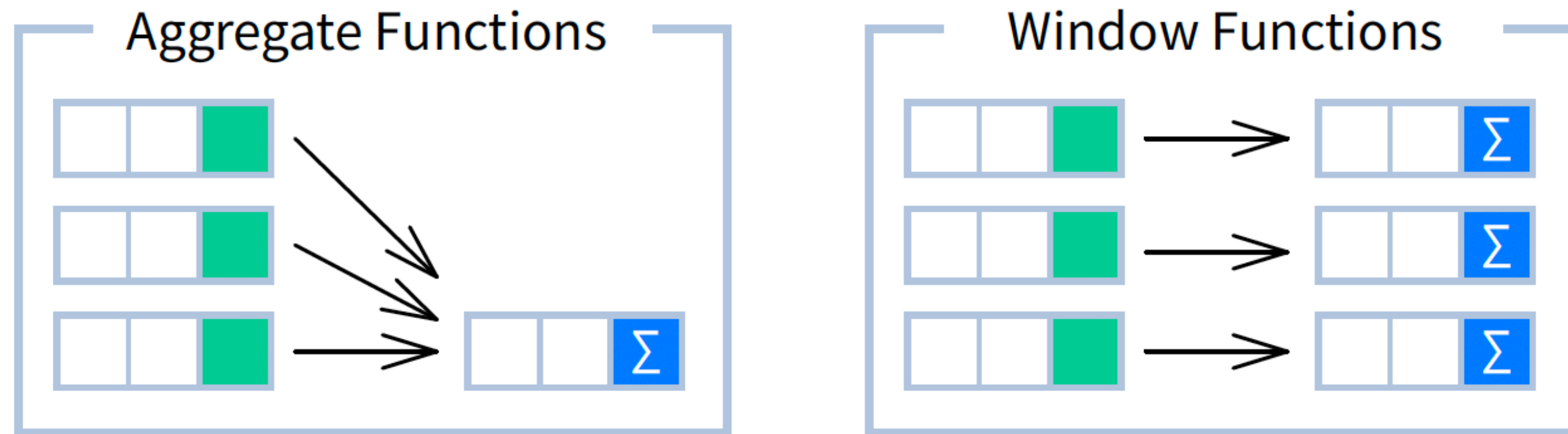
- AVG() Average value
- COUNT(), COUNT_BIG() Count of values
- MIN(), MAX() Minimum / Maximum of value
- SUM(), PRODUCT() Sum / product of values
- and also VAR(), VARP(), STDEV(), STDEVP(), CHECKSUM_AGG(int)

work as 'normal' aggregates, but also in conjunction with Window functions.



Aggregate functions

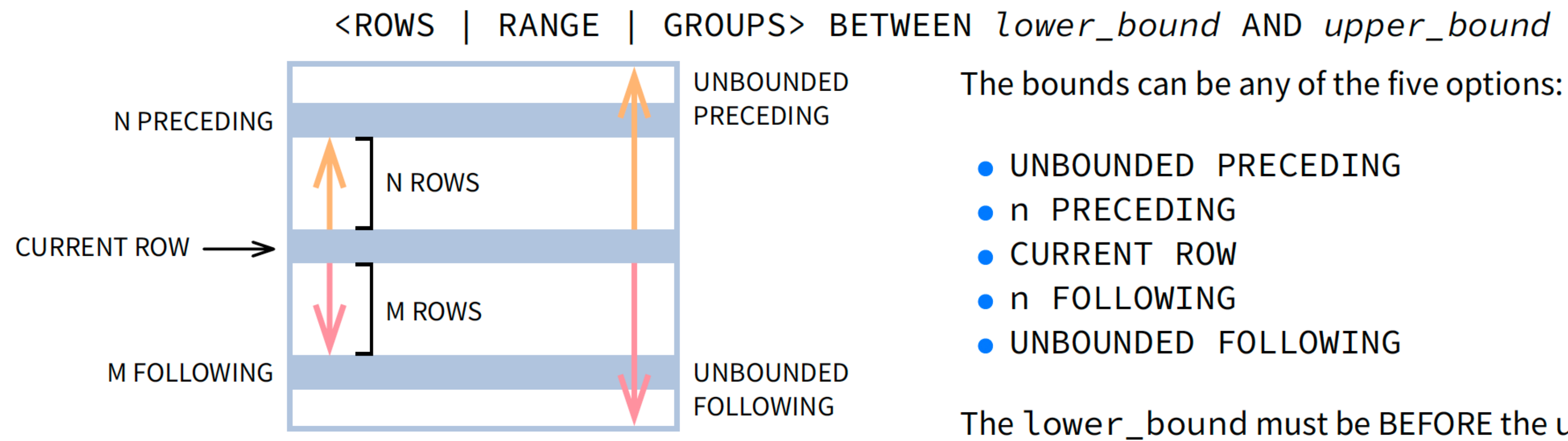
Grouped aggregates vs Window aggregates



Aggregating on a grouped level vs
aggregating on criteria relative to each individual row.

Window frames

- Introduce an additional level of filtering, relative to the current row, applicable to aggregate functions and First_value, Last_value, [Nth_value]



The default behaviour with an optional ORDER BY is:
RANGE BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW



Visualization by: <https://learnsql.com/blog/sql-window-functions-cheat-sheet/>

Ranking functions

- `ROW_NUMBER()`
returns the sequential number of each row within a partition
- `RANK()`
similar to `ROW_NUMBER()`, but returns the same value for ties, allows gaps
- `DENSE_RANK()`
returns the same rank value for ties, without gaps
- `NTILE(n)`
evenly distributes the rows into the specified number of groups,
returns the group number for each row; i.e. `NTILE(4)` -> quartiles



all of them require `OVER(ORDER BY ...)`



Statistical / analytical functions

Rank distribution:

- `PERCENT_RANK()`
calculates the relative rank of a row within a partition
- `CUME_DIST()`
returns the cumulative distribution of a row within a partition

Inverse distribution:

- `PERCENTILE_CONT`
calculates a percentile based on continuous distribution
- `PERCENTILE_DISC`
returns the percentile based on discrete distribution



Offset functions

- LAG(expr, offset, default) value from row before current row
- LEAD(expr, offset, default) value from row after current row
- FIRST_VALUE() by order within the partition (not the smallest value)
- LAST_VALUE() by order within the partition (not the highest value)
- [NTH_VALUE()] not implemented in SQL Server



Specials

- Ordered set functions
 - STRING_AGG() - in SQL-standard called LISTAGG
 - [Hypothetical set functions] grouped
 - [Inverse distribution functions] grouped
- [Row-pattern recognition] Totally not implemented in SQL Server 😞
- First and last record in a partition



Specials: Gaps and islands

GapStart

GapEnd

2025-09-24

2025-09-24

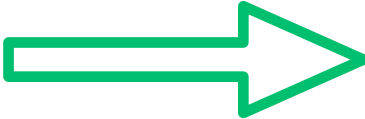
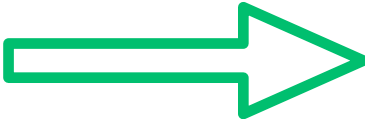
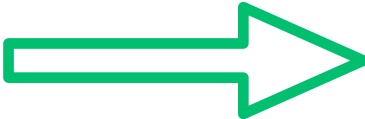
2025-09-27

2025-09-28

2025-10-03

2025-10-05

Gaps



2025-09-22

2025-09-23

2025-09-25

2025-09-26

2025-09-29

2025-09-30

2025-10-01

2025-10-02

2025-10-06

2025-10-07

2025-10-08

Islands

IslandStart

IslandEnd

2025-09-22

2025-09-23

2025-09-25

2025-09-26

2025-09-29

2025-10-02

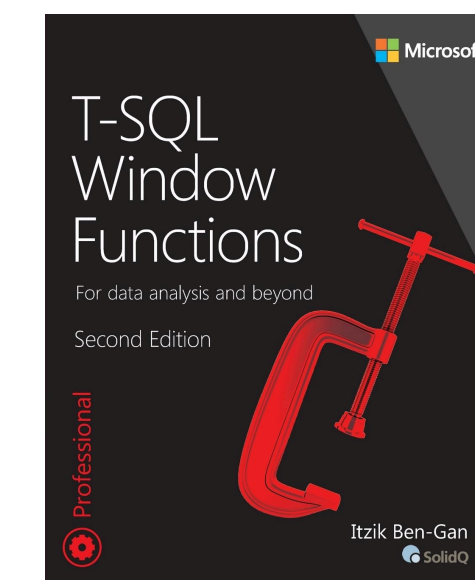
2025-10-06

2025-10-08



Resources:

- MS Learn: <https://learn.microsoft.com/en-us/training/modules/write-queries-that-use-window-functions/>
<https://learn.microsoft.com/en-us/sql/t-sql/queries/select-over-clause-transact-sql/>
<https://learn.microsoft.com/en-us/sql/t-sql/queries/select-window-transact-sql/>
- LearnSQL (incl. cheat sheet): <https://learnsql.com/blog/sql-window-functions-guide/>
- AW2025 download: <https://learn.microsoft.com/en-us/sql/samples/adventureworks-install-configure>
- AdventureWorks diagram: <https://www.dbdiagrams.com/online-diagrams/adventureworks/>
- Itzik Ben-Gan: „T-SQL Window Functions - For data analysis and beyond“, ISBN 978-0135861448



Window functions in SQL Server

Thank you for your time and interest & keep in touch:

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 <https://bsky.app/profile/derfredo.bsky.social>

 de.linkedin.com/in/derfredo

This file and the demo scripts can be found at:

<https://bit.ly/DerFredoRheinland2026>

